PROJECT REPORT

NETFLIX DATA ANALYSIS

A Complete End-to-End Data Analytics Project

Using Python | MySQL | Power BI

Submitted in partial fulfilment of the requirements
for the Award of the Degree

Bachelor of Technology
(Computer Science and Engineering)

Submitted By

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Roll No: 22CET0235
Academic Year: 2022-2026

Under the Guidance of
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CERTIFICATE

This is to certify that the project report titled "Netflix Data Analysis" has been prepared and submitted by Aayush Naithani, Roll No. 22CET0235, a student of MERI COLLEGE, under my guidance and supervision.

The project is based on real-world data analysis using Python, MySQL, and Power BI. The work is original and has not been submitted elsewhere for any degree or examination.

Faculty Guide's Signature: _____

Head of Department: _____

Date: _____

STUDENT DECLARATION

I, Aayush Naithani, student of B.Tech Computer Science, Roll No. 22CET0235, hereby declare that the project report titled "Netflix Data Analysis" is my own original work.

This report has been prepared for the purpose of academic submission only. The data used in this project has been taken from publicly available sources (Kaggle). All tools used — Python, MySQL, and Power BI — have been used solely for educational and analytical purposes.

I also confirm that this work has not been submitted elsewhere and that all the results shown in this report are genuine findings from the analysis I have personally performed.

Student Signature: _____

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me in completing this project successfully.

First, I am very thankful to my faculty guide, [Faculty Name], for providing constant guidance and encouragement throughout this project. Their expert suggestions helped me solve several technical problems and improve the quality of my analysis.

I am also grateful to the Head of Department, Gaurav Bhateja, for providing the necessary resources and a supportive environment for learning.

I would also like to thank the dataset contributors on Kaggle who made the Netflix dataset publicly available. Without that data, this project would not have been possible.

Finally, I thank my family and friends for their constant moral support and encouragement during the entire duration of this project.

Aayush Naithani
B.Tech Computer Science, 2026, 22CET0235

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ABSTRACT

Netflix is one of the largest streaming platforms in the world, with millions of users and thousands of titles available in different countries. Understanding what kind of content is available, from which countries, and how it has grown over time is very useful information for decision makers.

This project performs a complete end-to-end data analysis of Netflix's content library using three powerful tools — Python for data cleaning and exploration, MySQL for structured database querying, and Power BI for visual dashboards.

The dataset used in this project contains information about Netflix shows and movies such as title, type (Movie or TV Show), country, year of release, genre, rating, and duration. The raw dataset had several quality issues including missing values, incorrect data types, and duplicate records, all of which were handled during the pre-processing stage.

The key findings of this project include: identifying the most content-producing countries, understanding the trend of content addition over the years, finding the most common genres, and comparing the ratio of Movies vs TV Shows on the platform.

Chapter 1: Introduction

1.1 Background

Netflix, Inc. is an American subscription-based streaming service and production company. Founded in 1997 by Reed Hastings and Marc Randolph in Scotts Valley, California, it started as a DVD-by-mail service and later transitioned to online streaming in 2007.

Today, Netflix operates in over 190 countries and offers a wide library of films, television series, documentaries, and stand-up specials. As of recent reports, Netflix has over 260 million paid subscribers worldwide, making it one of the most popular entertainment platforms on the planet.

The massive scale of Netflix's content library makes it a very interesting subject for data analysis. There is a publicly available dataset on Kaggle that contains detailed information about Netflix's movies and TV shows, including their type, title, director, cast, country of origin, date added to the platform, release year, content rating, duration, genre, and description.

1.2 Why This Project?

In today's world, data is being generated at a very fast pace. Every time someone watches a movie, searches for a show, or gives a rating, data is created. Data Analytics helps organizations and individuals make sense of this data and draw useful conclusions from it.

This project was chosen because:

- Netflix data is widely available and relatable to everyday life.
- It covers the full data analytics pipeline — from raw data to visual insights.
- It provides hands-on experience with three major industry tools: Python, SQL, and Power BI.
- The analysis results are easy to understand and present.

1.3 Scope of the Project

This project covers the following areas of data analytics:

- Data Collection: Obtaining the Netflix dataset from Kaggle.
- Data Cleaning: Removing or fixing problems in the raw data using Python (Pandas library).
- Exploratory Data Analysis (EDA): Understanding the data by plotting basic charts and getting summary statistics.
- Database Integration: Loading the cleaned data into a MySQL database for structured querying.
- SQL Analysis: Writing SQL queries to answer specific business questions about the data.
- Data Visualization: Creating an interactive Power BI dashboard to display the insights visually.

This project does not involve machine learning or predictive modelling. It focuses purely on descriptive analytics — understanding what has already happened in Netflix's content library.

Chapter 2: Objectives of the Project

2.1 Main Objective

The main objective of this project is to analyse the Netflix content dataset and discover meaningful patterns and trends using Python, MySQL, and Power BI.

2.2 Specific Objectives

The following are the specific goals this project aims to achieve:

1. To understand the structure and quality of the Netflix dataset and prepare it for analysis.
2. To clean the dataset by handling missing values, incorrect data types, and duplicate records.
3. To perform Exploratory Data Analysis (EDA) and generate basic visual summaries.
4. To push the cleaned data into a MySQL database and run structured queries.
5. To find out which countries produce the most Netflix content.
6. To compare the number of Movies vs TV Shows on the platform.
7. To study the trend of content addition on Netflix over the years.
8. To identify the most popular genres on Netflix.
9. To create an interactive Power BI dashboard to present findings visually.

2.3 Expected Outcomes

By the end of this project, the following outcomes are expected:

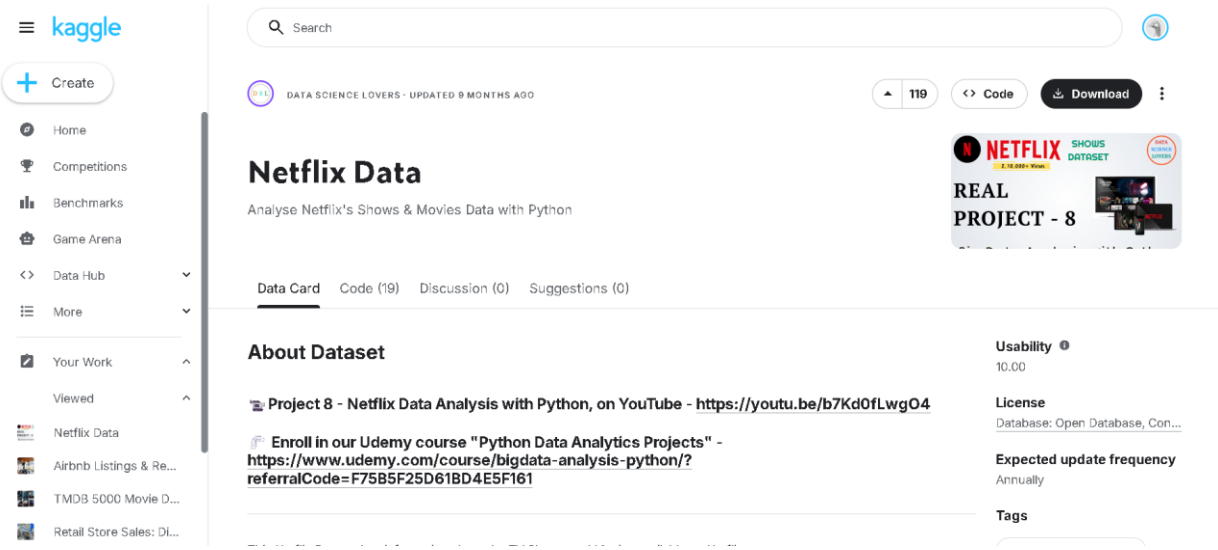
- A fully cleaned dataset ready for analysis.
- A set of meaningful SQL query results that answer business questions.
- A professional Power BI dashboard with multiple visualizations.
- A complete understanding of the end-to-end data analytics workflow.

Chapter 3: About the Dataset

3.1 Dataset Source

The dataset used in this project is the Netflix Movies and TV Shows dataset, which is freely available on Kaggle. The link to the dataset is: https://www.kaggle.com/datasets/shivamb/netflix-shows

This dataset was created by scraping Netflix's publicly accessible content library. It contains information about all the Movies and TV Shows available on Netflix up to a certain date.



3.2 Dataset Description

The dataset is in CSV format and contains approximately 8,800 records. Each record represents one title (either a Movie or a TV Show) on Netflix.

Below is a description of each column in the dataset:

Column Name	Data Type	Description
show_id	String	A unique ID for each title on Netflix
type	String	Whether the title is a 'Movie' or a 'TV Show'
title	String	The name of the Movie or TV Show
director	String	Name of the director (can be missing)
cast	String	Names of the main cast members (can be missing)
country	String	Country where the title was produced
date_added	String	Date the title was added to Netflix
release_year	Integer	The year the title was originally released
rating	String	Content rating (e.g., PG, TV-MA, R)
duration	String	Duration in minutes (for Movies) or number of Seasons (for TV Shows)
listed_in	String	Genre or category of the title
description	String	A short text description of the title

3.3 Data Quality Issues Found

Before cleaning, the dataset had several quality problems:

- Missing Values: The 'director', 'cast', 'country', and 'date_added' columns had a significant number of missing values.
- Wrong Data Type: The 'date_added' column was stored as a plain text string instead of a proper date format.
- Mixed Duration Format: The 'duration' column had different formats — '90 min' for Movies and '2 Seasons' for TV Shows.
- Duplicate Records: Some rows appeared more than once in the dataset.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7789 entries, 0 to 7788
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Show_Id       7789 non-null   object
 1   Category      7789 non-null   object
 2   Title         7789 non-null   object
 3   Director      5401 non-null   object
 4   Cast          7071 non-null   object
 5   Country       7282 non-null   object
 6   Release_Date  7779 non-null   object
 7   Rating        7782 non-null   object
 8   Duration      7789 non-null   object
 9   Type          7789 non-null   object
 10  Description   7789 non-null   object
dtypes: object(11)
memory usage: 669.5+ KB
```

```
[4]: df.isna().sum()
```

```
[4]: Show_Id            0
     Category           0
     Title              0
     Director        2388
     Cast             718
     Country          507
     Release_Date      10
     Rating             7
     Duration           0
     Type               0
     Description        0
     dtype: int64
```

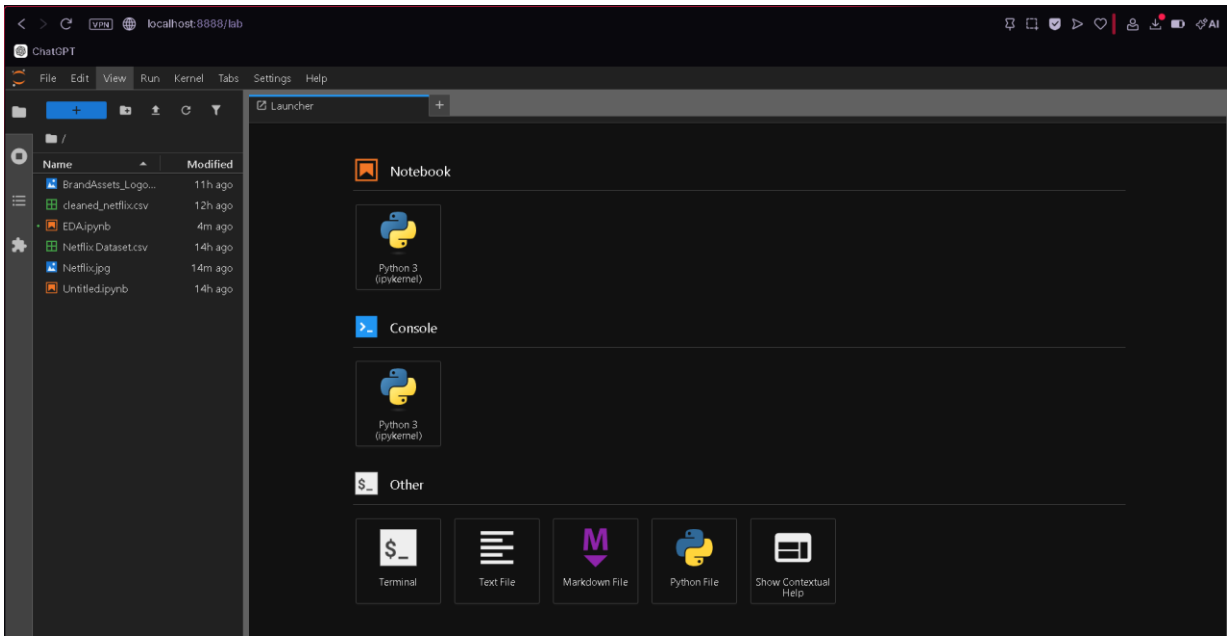
Chapter 4: Tools & Technologies Used

4.1 Python

Python is a high-level, general-purpose programming language that is widely used in data science and analytics. It was first released in 1991 by Guido van Rossum and has grown to become the most popular language for data-related tasks.

In this project, Python was used for data loading, cleaning, and exploratory data analysis. The following Python libraries were used:

Library	Purpose
Pandas	For loading, cleaning, and manipulating the dataset (DataFrames)
NumPy	For numerical operations and array handling
Matplotlib	For creating basic charts and plots during EDA
SQLAlchemy	For connecting Python to the MySQL database and uploading the cleaned data

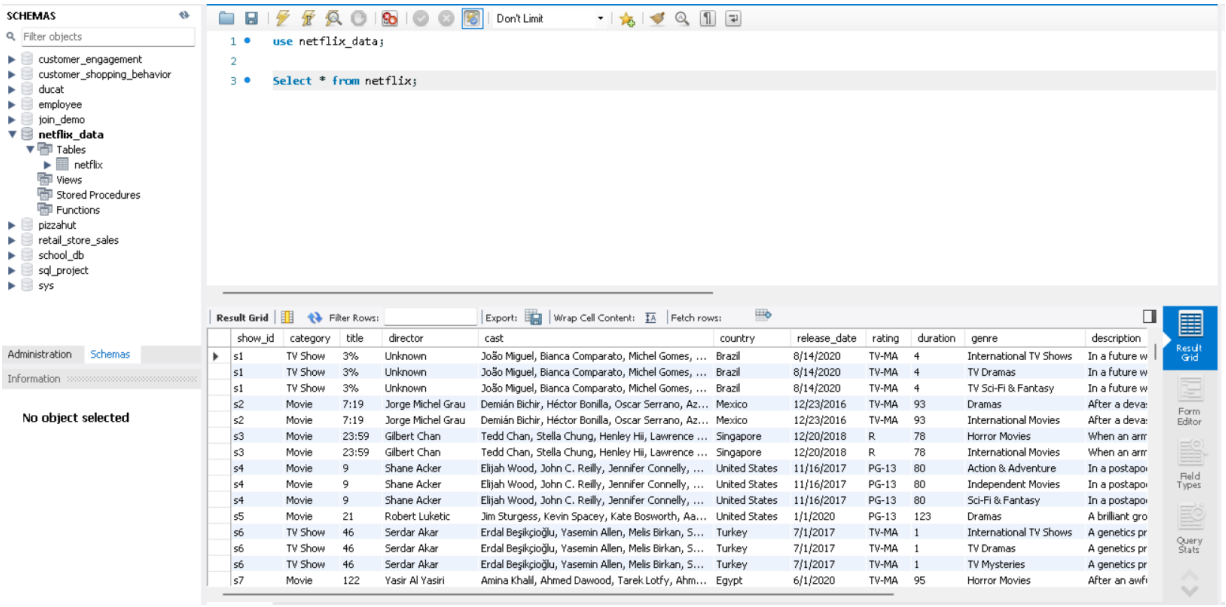


4.2 MySQL

MySQL is one of the most popular open-source relational database management systems (RDBMS) in the world. It uses SQL (Structured Query Language) to store, retrieve, and manage data in a structured, tabular format.

In this project, MySQL was used to:

- Store the cleaned Netflix dataset in a structured database.
- Run queries to answer specific questions about the data.
- Demonstrate real-world database operations such as grouping, counting, ordering, and filtering.

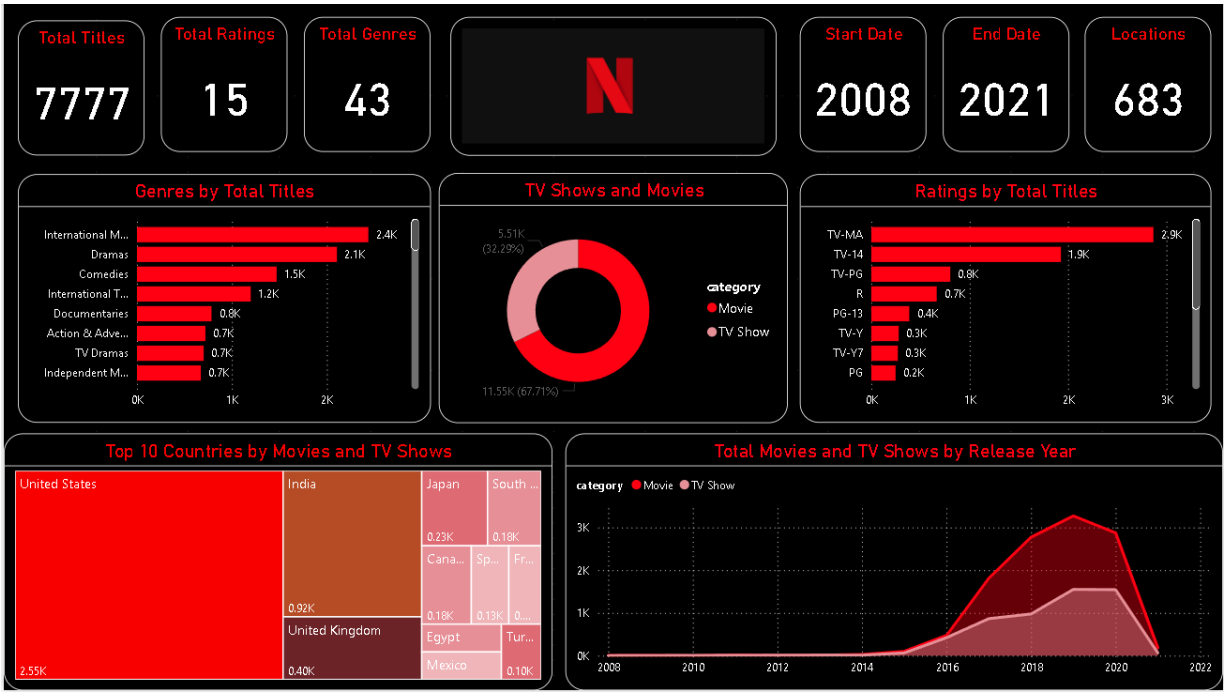


4.3 Power BI

Power BI is a business analytics tool developed by Microsoft. It allows users to connect to various data sources and create interactive dashboards and reports. It is widely used in the industry for data visualization and reporting.

In this project, Power BI was used to:

- Import the cleaned Netflix dataset.
- Create bar charts, pie charts, line charts, and card visuals.
- Build an interactive dashboard that displays all key insights in one place.



4.4 Jupyter Notebook

Jupyter Notebook is an open-source web-based application that allows developers and data analysts to write and run code in blocks (called 'cells'). It supports Python and is widely used for data science projects because it combines code, output, and explanations in a single document.

All Python code for this project was written and executed inside Jupyter Notebook, making it easy to run code step-by-step and see results immediately.

Chapter 5: Data Cleaning & Pre-Processing (Python / EDA)

5.1 Loading the Dataset

The first step of the project was to load the raw Netflix CSV dataset into Python using the Pandas library. The code used for this is shown below:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.read_csv("Netflix Dataset.csv")
```

After loading the file, the df.info() function was used to get an overview of the dataset — the number of rows and columns, the data type of each column, and how many non-null values each column contains.

5.2 Handling Missing Values

Several columns in the dataset had missing values. The following strategy was used to handle them:

Column	Problem	Solution Applied
director	Many missing values	Filled with the text 'Unknown'
cast	Many missing values	Filled with the text 'Unknown'
country	Some missing values	Filled with the text 'Unknown'
release_date	A few missing values	Rows with missing dates were dropped
rating	A few missing values	Filled with the most common (mode) rating value

```
df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df.dropna(subset=['release_date'], inplace=True)
df['rating'].fillna(df['rating'].mode()[0], inplace=True)
```

```
[27]: df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 7777 entries, 0 to 7788
Data columns (total 13 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       7777 non-null   object
 1   category      7777 non-null   object
 2   title         7777 non-null   object
 3   director      7777 non-null   object
 4   cast          7777 non-null   object
 5   country       7777 non-null   object
 6   release_date  7777 non-null   datetime64[ns]
 7   rating        7777 non-null   object
 8   duration      7777 non-null   object
 9   type          7777 non-null   object
 10  description   7777 non-null   object
 11  year_added    7777 non-null   int32
 12  month_added   7777 non-null   int32
dtypes: datetime64[ns](1), int32(2), object(10)
memory usage: 789.9+ KB
```

5.3 Converting Data Types

The 'release_added' column was stored as a plain text string. To perform time-based analysis, it needed to be converted into a proper DateTime format. Additionally, new columns for 'year_added' and 'month_added' were extracted from this date column.

```
df['release_date'] = pd.to_datetime(df['release_date'], format='%d-%m-%Y')
df['year_added'] = df['release_date'].dt.year
df['month_added'] = df['release_date'].dt.month
```

```
[3]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7789 entries, 0 to 7788
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Show_Id       7789 non-null   object
 1   Category      7789 non-null   object
 2   Title         7789 non-null   object
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 8   Duration      7789 non-null   object
 9   Type          7789 non-null   object
 10  Description   7789 non-null   object
dtypes: object(11)
memory usage: 669.5+ KB
```

[21]: df.head()

[21]:

	show_id	category	title	director	cast	country	release_date	rating	duration	type	description
0	s1	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	14-Aug-20	TV-MA	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...
1	s2	Movie	7:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	23-Dec-16	TV-MA	93	Dramas, International Movies	After a devastating earthquake hits Mexico Cit...
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence ...	Singapore	20-Dec-18	R	78	Horror Movies, International Movies	When an army recruit is found dead, his fellow...
3	s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly...	United States	16-Nov-17	PG-13	80	Action & Adventure, Independent Movies, Sci-Fi...	In a postapocalyptic world, rag-doll robots hi...
4	s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bosworth, Aar...	United States	1-Jan-20	PG-13	123	Dramas	A brilliant group of students become card-coun...

5.4 Cleaning the Duration Column

The 'duration' column had mixed values — it showed '90 min' for movies and '2 Seasons' for TV shows. The text parts were removed to keep only the numeric values, making it easier to analyse.

```
df["duration"] = df["duration"].str.replace(" min", "")
df["duration"] = df["duration"].str.replace(" Seasons", "")
df["duration"] = df["duration"].str.replace(" Season", "")
```

5.5 Removing Duplicates

Duplicate rows can cause errors in analysis by inflating counts and percentages. The following command was used to remove all duplicate records from the dataset:

```
df.drop_duplicates(inplace=True)
```

5.6 Saving the Cleaned Data

After completing all cleaning steps, the final cleaned dataset was saved as a new CSV file so that it could be loaded into MySQL.

```
df.to_csv("cleaned_netflix.csv", index=False)
```


```
df.to_csv("cleaned_netflix.csv", index=False)
```

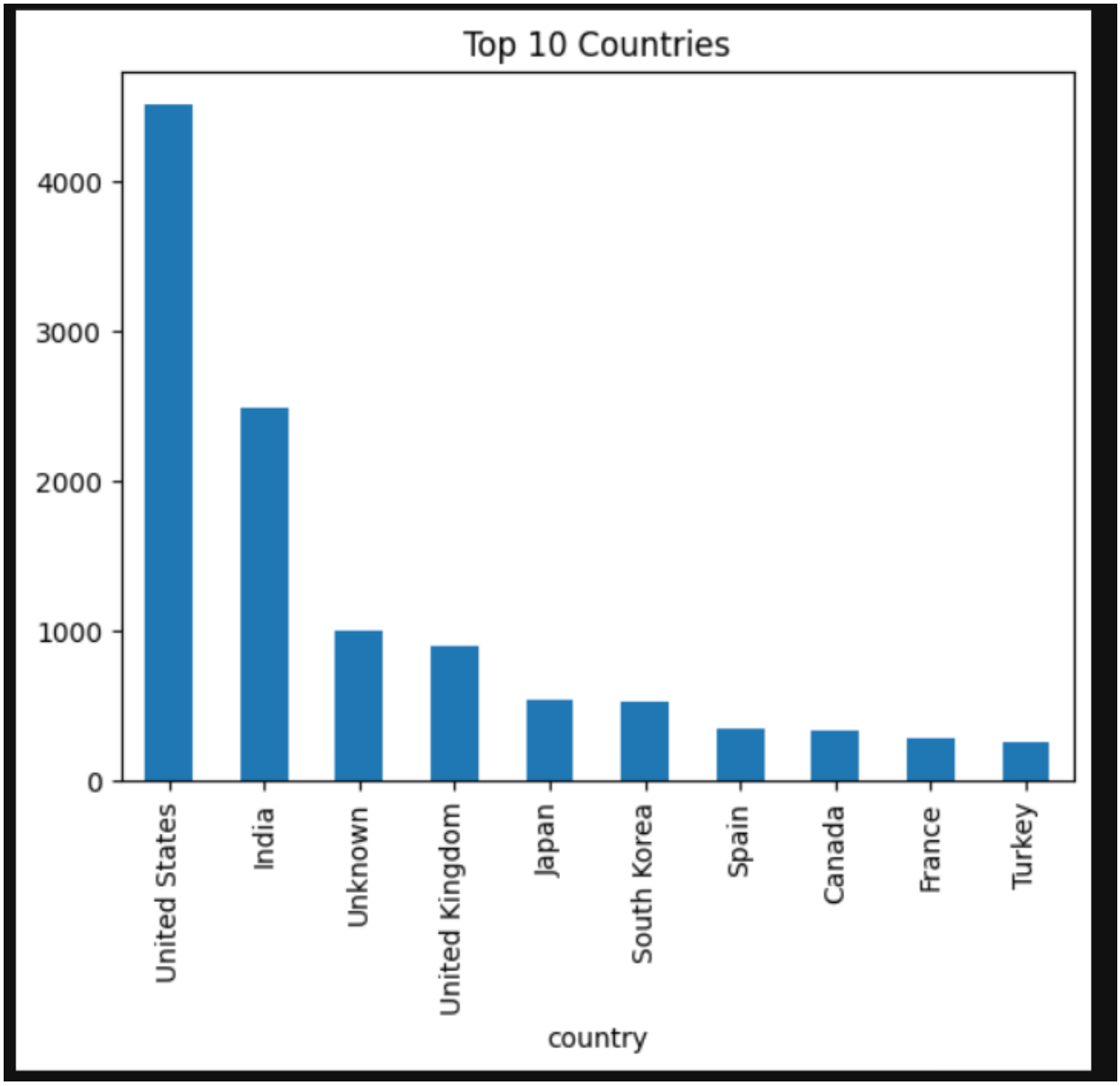

5.7 EDA Visualizations

After cleaning the data, basic exploratory analysis was done to understand the distribution and trends in the dataset. The following charts were created:

5.7.1 Top 10 Countries by Content

A bar chart was created to show which countries have produced the most content available on Netflix.

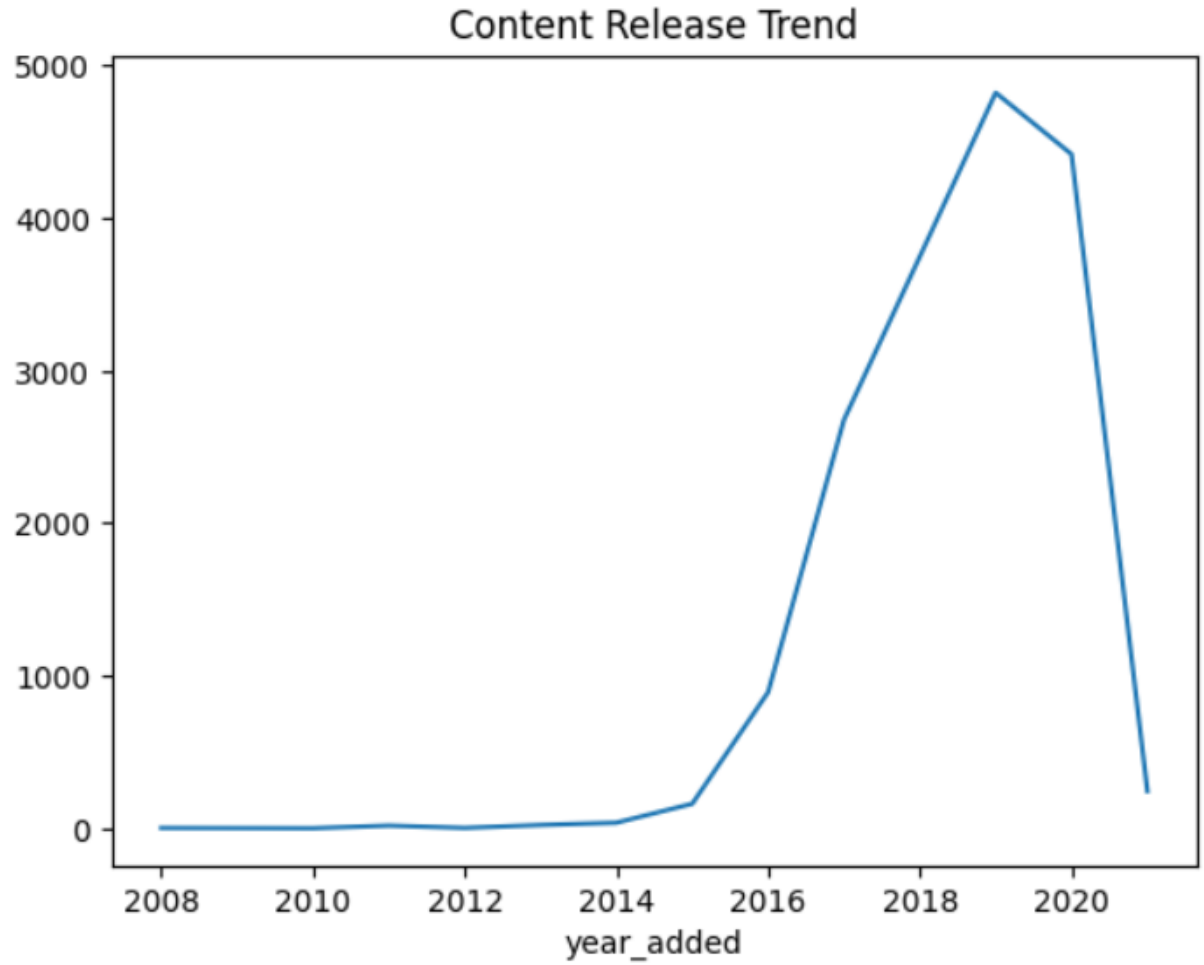
```
df['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Countries")
plt.show()q
```



5.7.2 Content Release Trend Over the Years

A line chart was created to show how the number of titles added to Netflix has changed over the years.

```
df['year_added'].value_counts().sort_index().plot()
plt.title("Content Release Trend")
plt.show()
```



Chapter 6: Loading Data into MySQL Database

6.1 Why Use a Database?

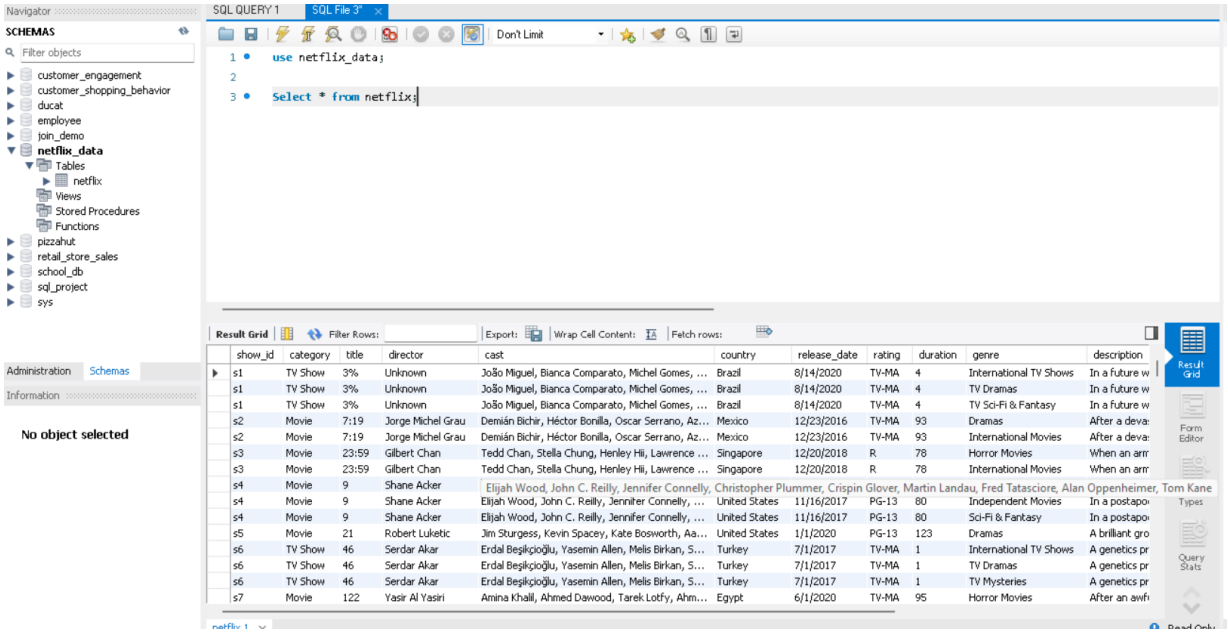
While a CSV file is useful for quick analysis in Python, a proper database like MySQL offers many advantages. It allows multiple people to access the data simultaneously, supports faster queries on large datasets, and provides a structured and organised way to store and retrieve information.

In real-world data analytics projects, data is almost always stored in databases. Therefore, this project simulates that professional workflow by pushing the cleaned data into MySQL.

6.2 Setting Up the MySQL Database

A new database called 'netflix_data' was created in MySQL Workbench. The following SQL command was used to create and select this database:

```
CREATE DATABASE netflix_data;
USE netflix_data;
```



6.3 Connecting Python to MySQL Using SQLAlchemy

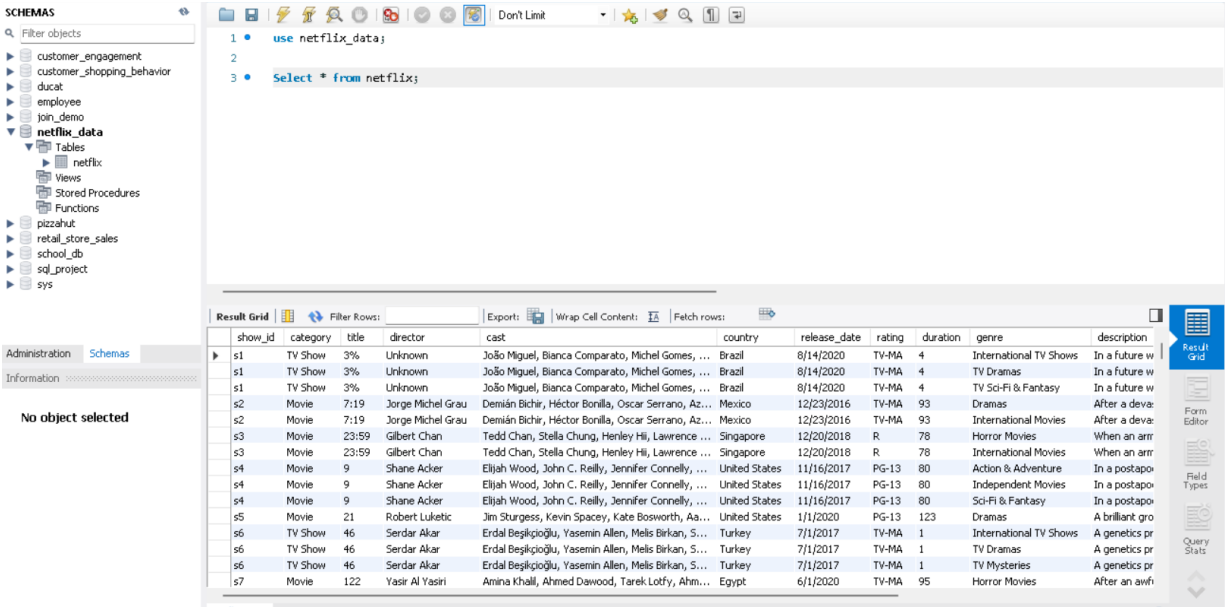
To transfer data from Python to MySQL automatically, the SQLAlchemy library was used. This library acts as a bridge between Python's Pandas library and the MySQL database. The connection string required the database username, password, host (localhost), port (3306), and database name.

```
from sqlalchemy import create_engine
engine =
create_engine("mysql+mysqldb://root:password@localhost:3306/netflix_data"
)
df.to_sql("netflix", engine, if_exists="replace", index=False)
print("Data inserted successfully")
```

The if_exists='replace' option tells SQLAlchemy to drop and recreate the table if it already exists, ensuring we always have fresh data.

```
# Pushing Refined Dataset into A DataBase

from sqlalchemy import create_engine

user = "root"
pw = "naithani%403"
host = "localhost"
port = "3306"
db = "netflix_data"

engine = create_engine(f"mysql+mysqldb://{user}:{pw}@{host}:{port}/{db}")

table_name = "netflix"
df.to_sql(table_name, engine, if_exists ="replace",index = False)
print("Data inserted succesfully")
```

```
Data inserted succesfully
```



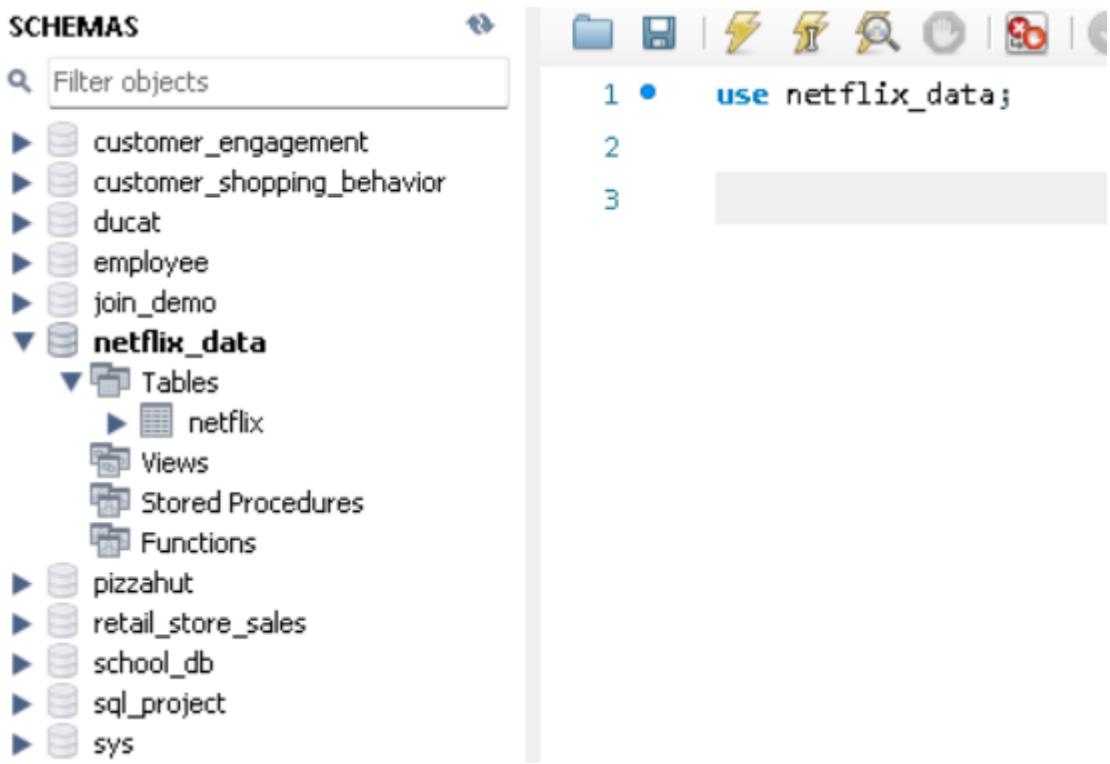
Chapter 7: SQL Queries & Analysis

7.1 Overview

After the cleaned Netflix data was loaded into the MySQL database, SQL (Structured Query Language) was used to run queries and extract meaningful insights. SQL is ideal for answering specific business questions from large structured datasets.

All queries were run inside MySQL Workbench. The database used was 'netflix_data' and the table used was 'netflix'.

```
USE netflix_data;
```



7.2 Query 1 – Movies vs TV Shows

This query counts how many titles in the database are Movies and how many are TV Shows.

```
SELECT genre, COUNT(*) FROM netflix GROUP BY genre;
```

This query groups all records by the 'genre' column (which contains type information — Movie or TV Show) and counts the number of entries in each group. The result shows us the basic split between Movies and TV Shows on Netflix.

genre	COUNT(*)
International TV Shows	1198
TV Dramas	703
TV Sci-Fi & Fantasy	75
Dramas	2106
International Movies	2437
Horror Movies	312
Action & Adventure	721
Independent Movies	673
Sci-Fi & Fantasy	218
TV Mysteries	90
Thrillers	491
Crime TV Shows	426
Docuseries	352
Documentaries	786
Sports Movies	196
Comedies	1471

7.3 Query 2 – Top 10 Countries by Content

This query finds the top 10 countries that have produced the most content available on Netflix.

```
SELECT country, COUNT(*) AS total
FROM netflix
GROUP BY country
ORDER BY total DESC
LIMIT 10;
```

The GROUP BY clause groups all records by country. The COUNT(*) function counts how many titles each country has. The ORDER BY total DESC clause sorts the results from highest to lowest, and LIMIT 10 shows only the top 10 results.

country	total
United States	4511
India	2496
Unknown	1009
United Kingdom	901
Japan	538
South Korea	524
Spain	356
Canada	339
France	286
Turkey	266

7.4 Query 3 – Content Added Per Year

This query shows how many new titles were added to Netflix each year, helping us understand the platform's content growth trend.

```
SELECT year_added, COUNT(*)
FROM netflix
GROUP BY year_added;
```

The result shows one row per year with the count of titles added in that year. This is useful for understanding whether Netflix has been growing its content library rapidly or slowly.

year_added	COUNT(*)
2020	4416
2016	893
2018	3746
2017	2677
2019	4819
2021	245
2015	162
2013	23
2011	19
2012	4
2014	38
2008	4
2009	3
2010	2

7.5 Query 4 – Most Common Genres

This query finds the top 10 most common genres in the Netflix library, helping understand which categories of content are most prevalent on the platform.

```
SELECT genre, COUNT(*) AS total
FROM netflix
GROUP BY genre
ORDER BY total DESC
LIMIT 10;
```

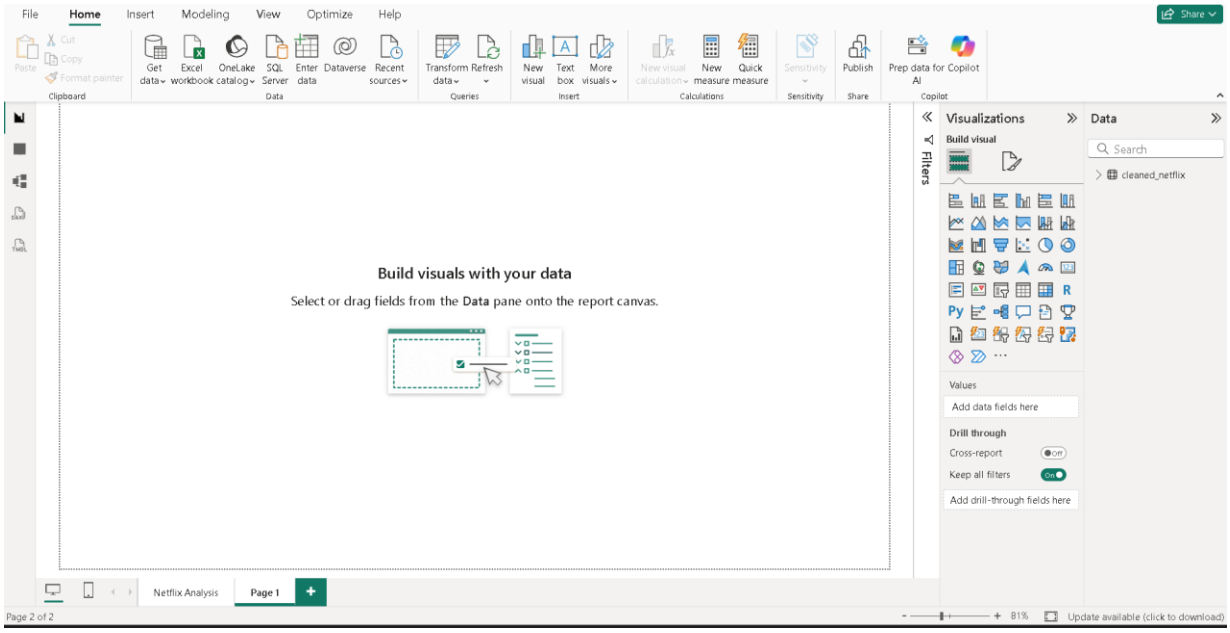
genre	total
International Movies	2437
Dramas	2106
Comedies	1471
International TV Shows	1198
Documentaries	786
Action & Adventure	721
TV Dramas	703
Independent Movies	673
Children & Family Movies	532
Romantic Movies	531

Chapter 8: Power BI Dashboard & Visualizations

8.1 Introduction to the Dashboard

After completing the Python data cleaning and MySQL SQL analysis, the final step was to create a visual and interactive dashboard using Microsoft Power BI. A dashboard makes it easy for anyone — even non-technical people — to understand the data insights at a glance.

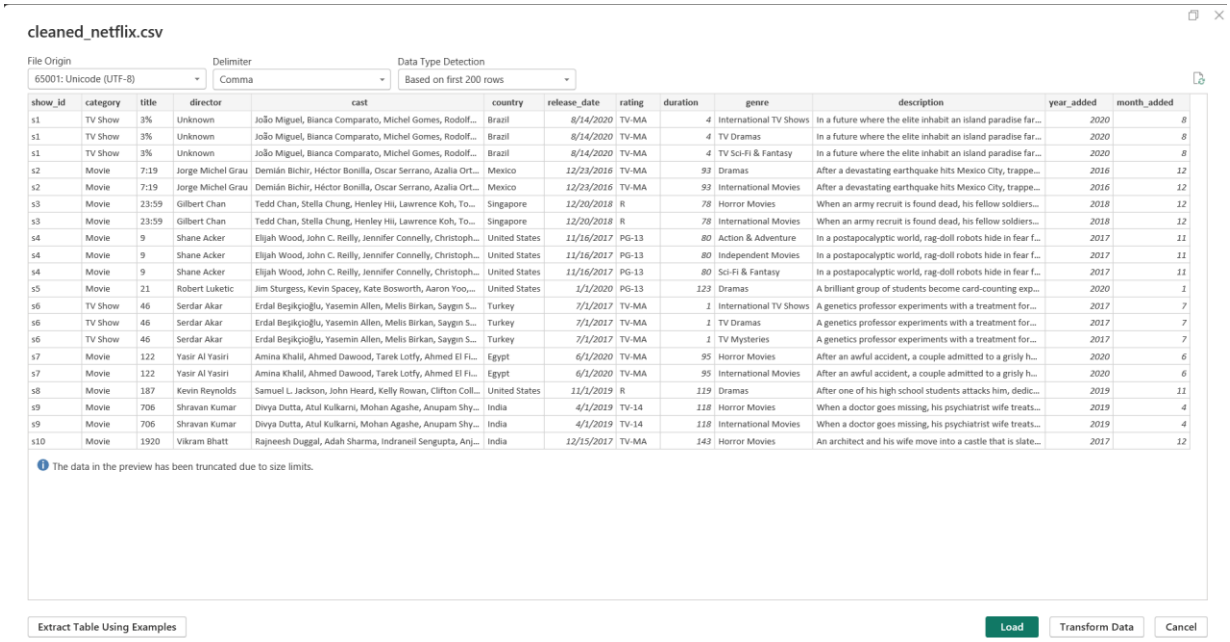
Power BI allows us to connect directly to files or databases, build visuals, and publish interactive reports. For this project, the cleaned Netflix CSV file was imported into Power BI Desktop.



8.2 Importing Data into Power BI

The cleaned dataset (cleaned_netflix.csv) was loaded into Power BI using the 'Get Data' option. Power BI automatically detected the column names and data types.

After importing, the Power Query editor was used to verify the data and make minor adjustments if needed. Once the data looked correct, it was loaded into the Power BI data model.



show_id	category	title	director	cast	country	release_date	rating	duration	genre	description	year_added	month_added
s1	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, Rodolf...	Brazil	8/14/2020	TV-MA	4	International TV Shows	In a future where the elite inhabit an island paradise far...	2020	8
s1	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, Rodolf...	Brazil	8/14/2020	TV-MA	4	TV Dramas	In a future where the elite inhabit an island paradise far...	2020	8
s1	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, Rodolf...	Brazil	8/14/2020	TV-MA	4	TV Sci-Fi & Fantasy	In a future where the elite inhabit an island paradise far...	2020	8
s2	Movie	7:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, Azalia Ort...	Mexico	12/23/2016	TV-MA	93	Dramas	After a devastating earthquake hits Mexico City, trappe...	2016	12
s2	Movie	7:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, Azalia Ort...	Mexico	12/23/2016	TV-MA	93	International Movies	After a devastating earthquake hits Mexico City, trappe...	2016	12
s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence Koh, To...	Singapore	12/20/2018	R	78	Horror Movies	When an army recruit is found dead, his fellow soldiers...	2018	12
s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence Koh, To...	Singapore	12/20/2018	R	78	International Movies	When an army recruit is found dead, his fellow soldiers...	2018	12
s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly, Christoph...	United States	11/16/2017	PG-13	80	Action & Adventure	In a postapocalyptic world, rag-doll robots hide in fear f...	2017	11
s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly, Christoph...	United States	11/16/2017	PG-13	80	Independent Movies	In a postapocalyptic world, rag-doll robots hide in fear f...	2017	11
s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly, Christoph...	United States	11/16/2017	PG-13	80	Sci-Fi & Fantasy	In a postapocalyptic world, rag-doll robots hide in fear f...	2017	11
s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bosworth, Aaron Yoo,...	United States	1/1/2020	PG-13	123	Dramas	A brilliant group of students become card-counting exp...	2020	1
s6	TV Show	46	Serdar Akar	Erdal Beşikçioğlu, Yasemin Allen, Melis Birkan, Saygın S...	Turkey	7/1/2017	TV-MA	1	International TV Shows	A genetics professor experiments with a treatment for...	2017	7
s6	TV Show	46	Serdar Akar	Erdal Beşikçioğlu, Yasemin Allen, Melis Birkan, Saygın S...	Turkey	7/1/2017	TV-MA	1	TV Dramas	A genetics professor experiments with a treatment for...	2017	7
s6	TV Show	46	Serdar Akar	Erdal Beşikçioğlu, Yasemin Allen, Melis Birkan, Saygın S...	Turkey	7/1/2017	TV-MA	1	TV Mysteries	A genetics professor experiments with a treatment for...	2017	7
s7	Movie	122	Yasir Al Yasiri	Amina Khalil, Ahmed Dawood, Tarek Lotfy, Ahmed El Fi...	Egypt	6/1/2020	TV-MA	95	Horror Movies	After an awful accident, a couple admitted to a grisly h...	2020	6
s7	Movie	122	Yasir Al Yasiri	Amina Khalil, Ahmed Dawood, Tarek Lotfy, Ahmed El Fi...	Egypt	6/1/2020	TV-MA	95	International Movies	After an awful accident, a couple admitted to a grisly h...	2020	6
s8	Movie	187	Kevin Reynolds	Samuel L. Jackson, John Heard, Kelly Rowan, Clifton Coll...	United States	11/1/2019	R	119	Dramas	After one of his high school students attacks him, dedic...	2019	11
s9	Movie	706	Shravan Kumar	Divya Dutta, Atul Kulkarni, Mohan Agashe, Anupam Shy...	India	4/1/2019	TV-14	118	Horror Movies	When a doctor goes missing, his psychiatrist wife treats...	2019	4
s9	Movie	706	Shravan Kumar	Divya Dutta, Atul Kulkarni, Mohan Agashe, Anupam Shy...	India	4/1/2019	TV-14	118	International Movies	When a doctor goes missing, his psychiatrist wife treats...	2019	4
s10	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengupta, Anj...	India	12/15/2017	TV-MA	143	Horror Movies	An architect and his wife move into a castle that is slate...	2017	12

8.3 Visuals Created

The following visuals were created in the Power BI dashboard:

8.3.1 KPI Cards

Three KPI (Key Performance Indicator) cards were placed at the top of the dashboard to show the most important numbers at a glance:

- Total Number of Titles on Netflix
- Total Number of Movies
- Total Number of TV Shows

8.3.2 Pie Chart – Movies vs TV Shows

A pie chart was created to visually show the proportion of Movies vs TV Shows in the Netflix library. This makes it immediately clear that Movies make up a larger share of the content than TV Shows.

8.3.3 Bar Chart – Top 10 Countries

A horizontal bar chart was created to display the top 10 countries by number of Netflix titles. Each bar represents one country, and the length of the bar shows the number of titles from that country.

8.3.4 Line Chart – Content Added Per Year

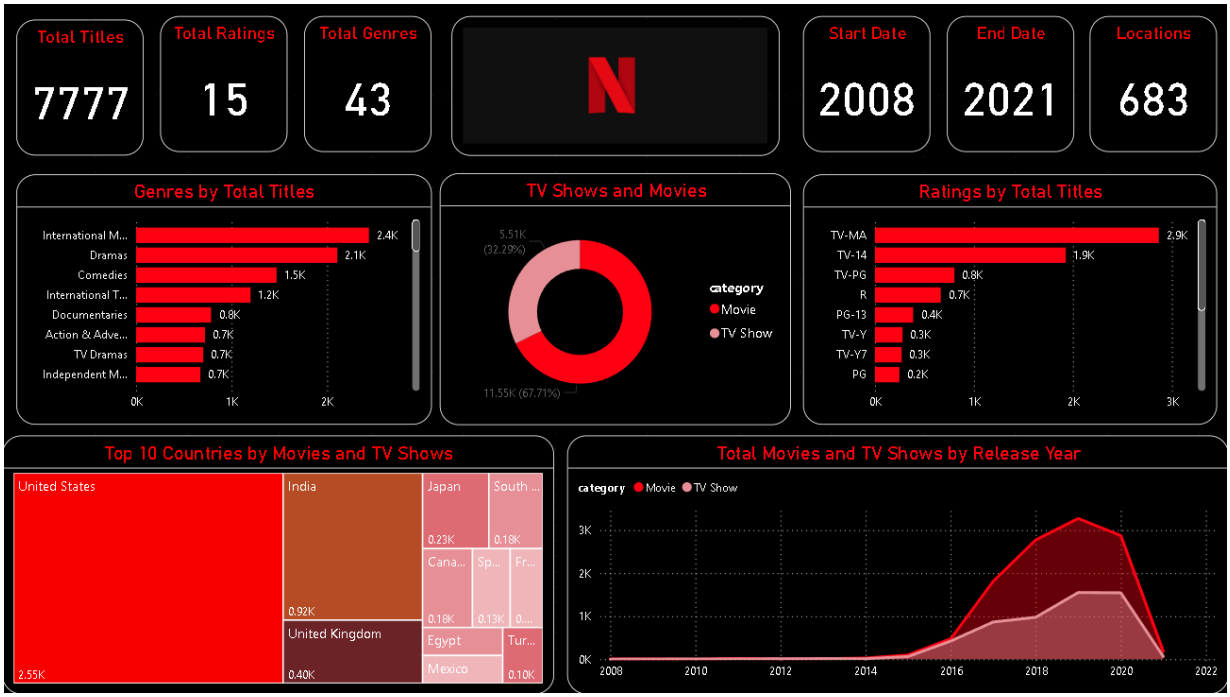
A line chart was created to show the growth trend of Netflix content over the years. The X-axis shows the year and the Y-axis shows the number of titles added in that year.

8.3.5 Bar Chart – Top 10 Genres

A bar chart was created to show the most common genres in Netflix's content library. This helps identify what kind of content is most available on the platform.

8.4 Final Dashboard Overview

All the above visuals were arranged together in a single Power BI report page to create a complete and interactive dashboard. Slicers were added to allow users to filter the data by year or by content type (Movie or TV Show).



Chapter 9: Key Findings & Insights

9.1 Movies vs TV Shows

One of the most basic but important findings from this analysis is that the majority of Netflix's content is Movies, not TV Shows. Approximately 70% of the titles in the dataset are Movies, while only 30% are TV Shows. This tells us that Netflix has historically focused more on offering movies rather than series.

genre	COUNT(*)
International TV Shows	1198
TV Dramas	703
TV Sci-Fi & Fantasy	75
Dramas	2106
International Movies	2437
Horror Movies	312
Action & Adventure	721
Independent Movies	673
Sci-Fi & Fantasy	218
TV Mysteries	90
Thrillers	491
Crime TV Shows	426
Docuseries	352
Documentaries	786
Sports Movies	196
Comedies	1471
Anime Series	147
Reality TV	222
TV Comedies	518
Romantic Movies	531
Romantic TV Shows	333
Science & Nature TV	85
Movies	56
British TV Shows	231
Korean TV Shows	150

9.2 Top Content Producing Countries

The analysis clearly shows that the United States is the top country in terms of Netflix content, with a very large number of titles compared to any other country. India comes in second place, followed by the United Kingdom.

This finding reflects the global dominance of the American entertainment industry, as well as Netflix's strong and growing focus on the Indian market, which is one of the largest consumer bases for streaming in the world.

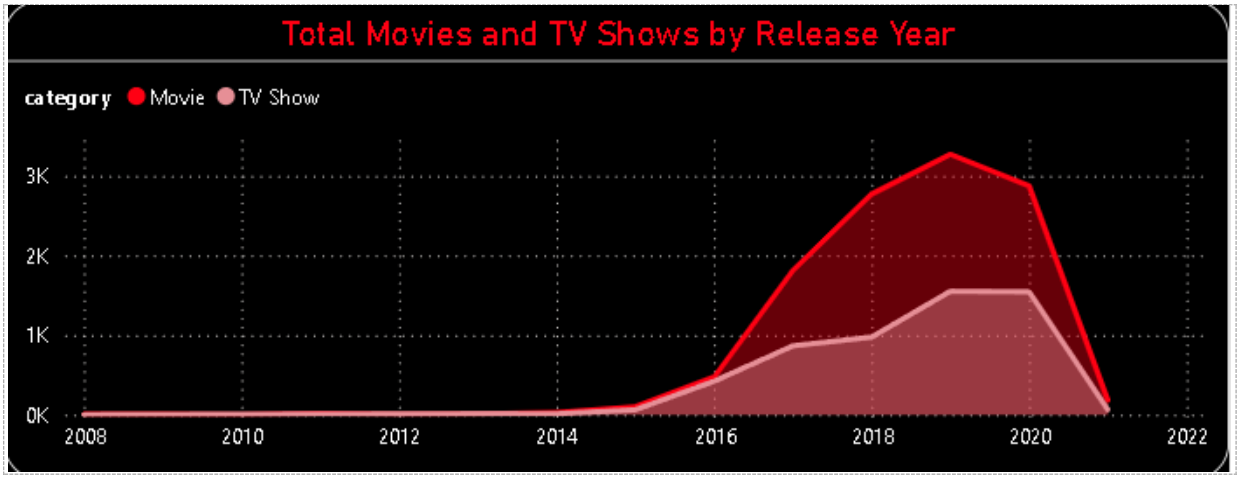
Rank	Country	Approx. Number of Titles
1	United States	~2800+
2	India	~900+
3	United Kingdom	~400+
4	Japan	~200+
5	South Korea	~150+

country	total
United States	4511
India	2496
Unknown	1009
United Kingdom	901
Japan	538
South Korea	524
Spain	356
Canada	339
France	286
Turkey	266

9.3 Content Growth Over the Years

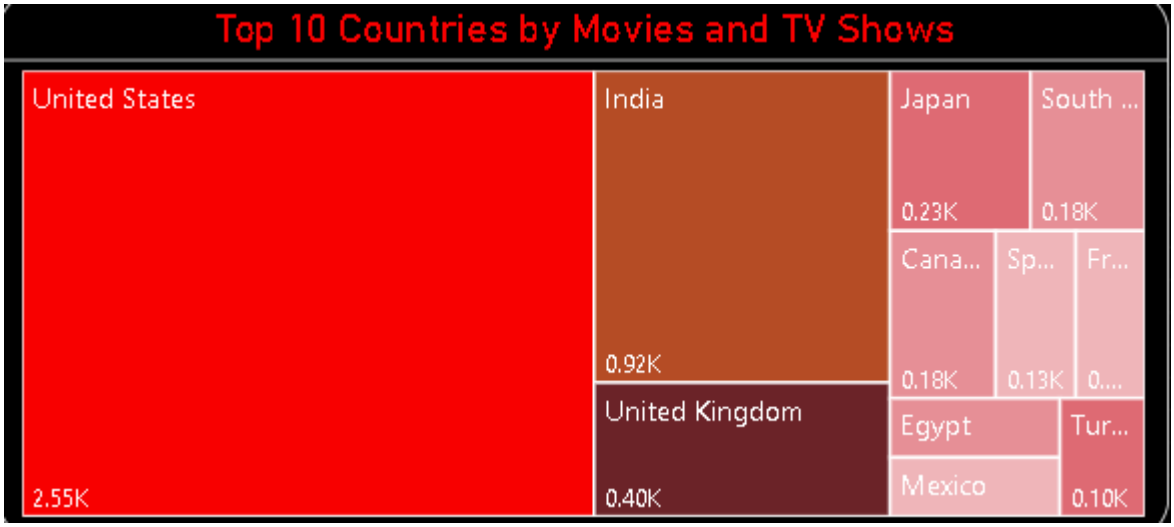
The year-wise analysis shows that Netflix has consistently grown its content library over time. Content additions were relatively slow in the early years (before 2015), but there was a sharp increase from 2016 to 2019. This was the period when Netflix began expanding aggressively into international markets.

After 2019, there was a slight slowdown, possibly due to the COVID-19 pandemic which affected film production globally. However, content additions picked back up in subsequent years.



9.4 Most Popular Genres

The genre analysis reveals that 'International Movies' and 'Dramas' are the most common genres on Netflix. This is followed by 'Comedies', 'Action & Adventure', and 'Documentaries'. The popularity of International Movies as a genre indicates Netflix's strong push to offer global content to its international subscriber base.



Chapter 10: Challenges Faced

10.1 Data Quality Issues

The raw dataset had many missing values, especially in the 'director', 'cast', and 'country' columns. Deciding on the right strategy for each column — whether to fill with a placeholder, use the mode, or drop the row — required careful thought. Incorrect handling could have led to biased or misleading results.

10.2 Data Type Conversion

The 'date_added' column was in a string format, and converting it to a proper date format required specifying the exact format string. An incorrect format would have caused errors or produced wrong year/month values. Several trials were required to get the right format ('%d-%m-%Y') working correctly.

10.3 MySQL Connection Setup

Setting up the connection between Python and MySQL using SQLAlchemy was tricky. Errors related to the wrong password encoding, incorrect port, or missing MySQL connector library had to be debugged carefully. The special characters in the password (like '@') needed to be URL-encoded to avoid connection failures.

10.4 Power BI Data Refresh

When changes were made to the CSV file, the Power BI dashboard did not always update automatically. The 'Refresh' button had to be clicked manually to reload the data. Setting up automatic data refresh required connecting to a live source, which was beyond the scope of this project.

10.5 Genre Column Complexity

The 'listed_in' column (genre) in the original dataset often contained multiple genres separated by commas (for example, 'Dramas, International Movies, Romantic Movies'). For the purpose of this project, the genre column was analysed as-is without splitting individual genre tags. A more advanced analysis would require exploding this column into individual rows.

Chapter 11: Conclusion

This project successfully demonstrated a complete end-to-end data analytics workflow using real-world Netflix data. Starting from a raw, messy CSV file and ending with a clean, interactive Power BI dashboard, every stage of the data analytics pipeline was covered.

The data cleaning stage in Python removed all null values, fixed data types, and eliminated duplicate records. The SQL queries in MySQL answered specific business questions and extracted structured insights. The Power BI dashboard brought all the findings together in a visually appealing and easy-to-understand format.

The key insights from this project are:

- Movies make up a larger proportion of Netflix's content compared to TV Shows.
- The United States is the dominant content producer on Netflix, followed by India and the United Kingdom.
- Netflix's content library grew rapidly between 2016 and 2019, reflecting its global expansion strategy.
- International Movies, Dramas, and Comedies are the most prevalent genres on the platform.

This project was not only a technical exercise but also a practical learning experience. It provided hands-on knowledge of three of the most widely used tools in data analytics — Python, SQL, and Power BI — and demonstrated how they can be combined to solve real-world analytical problems.

Chapter 12: Future Scope

While this project covered the essential aspects of data analytics, there are many ways in which it can be extended in the future:

12.1 Machine Learning for Recommendations

A recommendation system could be built on top of this dataset. By analysing genres, ratings, and viewing trends, a machine learning model could suggest movies or TV shows to users based on their preferences. This would add a predictive analytics layer to the current descriptive analytics project.

12.2 Sentiment Analysis on Descriptions

The 'description' column contains short text descriptions for each title. Natural Language Processing (NLP) techniques like sentiment analysis could be applied to this column to understand the tone and themes of Netflix content.

12.3 Real-Time Data Integration

This project used a static dataset downloaded from Kaggle. In the future, the Netflix API (if made available) or web scraping techniques could be used to automatically update the dataset with new titles as they are added to the platform, making the Power BI dashboard a live, always-updated report.

12.4 More Advanced SQL Analysis

More complex SQL queries can be written — for example, finding directors with the most titles, identifying the most prolific actors on the platform, or analysing seasonal trends in content addition (which months see the most new titles).

12.5 Cross-Platform Comparison

A future project could compare Netflix's content library with those of other streaming platforms such as Amazon Prime Video, Disney+, or Hotstar. This would provide competitive intelligence about which platform has more content in different categories and countries.

Chapter 13: References

The following sources were used during the preparation of this project:

10. Netflix Movies and TV Shows Dataset – Kaggle. Available at: https://www.kaggle.com/datasets/shivamb/netflix-shows
11. Python Pandas Documentation – https://pandas.pydata.org/docs/
12. Python Matplotlib Documentation – https://matplotlib.org/stable/contents.html
13. SQLAlchemy Documentation – https://docs.sqlalchemy.org/
14. MySQL 8.0 Reference Manual – https://dev.mysql.com/doc/refman/8.0/en/
15. Microsoft Power BI Documentation – https://learn.microsoft.com/en-us/power-bi/
16. NumPy Documentation – https://numpy.org/doc/
17. Jupyter Notebook Documentation – https://jupyter-notebook.readthedocs.io/
18. Netflix Wikipedia Page – https://en.wikipedia.org/wiki/Netflix
19. Data Cleaning Best Practices – Towards Data Science (https://towardsdatascience.com)

Chapter 14: Appendix

Appendix A – Complete Python Code

Below is the complete Python code used in this project for data loading, cleaning, and uploading to MySQL:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sqlalchemy import create_engine

# Load dataset
df = pd.read_csv("Netflix Dataset.csv")

# Standardise column names
df.columns = df.columns.str.strip().str.lower()

# Handle missing values
df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df.dropna(subset=['release_date'], inplace=True)
df['rating'].fillna(df['rating'].mode()[0], inplace=True)

# Convert date column
df['release_date'] = pd.to_datetime(df['release_date'], format='%d-%m-%Y')
df['year_added'] = df['release_date'].dt.year
df['month_added'] = df['release_date'].dt.month

# Clean duration column
df["duration"] = df["duration"].str.replace(" min", "")
df["duration"] = df["duration"].str.replace(" Seasons", "")
df["duration"] = df["duration"].str.replace(" Season", "")

# Remove duplicates
df.drop_duplicates(inplace=True)

# Save cleaned CSV
df.to_csv("cleaned_netflix.csv", index=False)

# Upload to MySQL
engine =
create_engine("mysql+mysqldb://root:password@localhost:3306/netflix_data")
df.to_sql("netflix", engine, if_exists="replace", index=False)
print("Data inserted successfully")
```

Appendix B – Complete SQL Queries

Below are all the SQL queries used in this project:

```
USE netflix_data;

-- Movies VS Shows
SELECT genre, COUNT(*) FROM netflix GROUP BY genre;

-- Top Countries
SELECT country, COUNT(*) AS total
```

```
FROM netflix
GROUP BY country
ORDER BY total DESC
LIMIT 10;

-- Content Added Per Year
SELECT year_added, COUNT(*)
FROM netflix
GROUP BY year_added;

-- Most Common Genres
SELECT genre, COUNT(*) AS total
FROM netflix
GROUP BY genre
ORDER BY total DESC
LIMIT 10;
```

Appendix C – Tools & Software Versions

Tool / Software	Version Used
Python	3.10+
Pandas	1.5+
NumPy	1.23+
Matplotlib	3.6+
SQLAlchemy	1.4+
MySQL	8.0
MySQL Connector	mysql-connector-python
Power BI Desktop	Latest (2024)
Jupyter Notebook	6.5+
Operating System	Windows 10/11